

Program : Computer Engineering Program Date : **12 April 2026**
Examiners : Dr. Tarek salem Duration **2 hours**
Eng : Eman, Habiba, Basant
Course/Code : Database Management Systems /
CSE112

Create database Company:-

1.

```
CREATE TABLE EMPLOYEE
( Fname          VARCHAR(15)          NOT NULL,
  Minit          CHAR,
  Lname          VARCHAR(15)          NOT NULL,
  Ssn            CHAR(9)              NOT NULL,
  Bdate          DATE,
  Address        VARCHAR(30),
  Sex            CHAR,
  Salary         DECIMAL(10,2),
  Super_ssn      CHAR(9),
  Dno            INT                  NOT NULL,
  PRIMARY KEY (Ssn);
```

2.

```
CREATE TABLE DEPARTMENT
( Dname          VARCHAR(15)          NOT NULL,
  Dnumber        INT                  NOT NULL,
  Mgr_ssn        CHAR(9)              NOT NULL,
  Mgr_start_date DATE,
  PRIMARY KEY (Dnumber),
  UNIQUE (Dname),
  FOREIGN KEY (Mgr_ssn) REFERENCES EMPLOYEE(Ssn) );
```

3.

```
CREATE TABLE DEPT_LOCATIONS
( Dnumber        INT                  NOT NULL,
  Dlocation      VARCHAR(15)          NOT NULL,
  PRIMARY KEY (Dnumber, Dlocation),
  FOREIGN KEY (Dnumber) REFERENCES DEPARTMENT(Dnumber) );
```

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4.

CREATE TABLE PROJECT

(Pname	VARCHAR(15)	NOT NULL,
Pnumber	INT	NOT NULL,
Plocation	VARCHAR(15),	
Dnum	INT	NOT NULL,

PRIMARY KEY (Pnumber),
UNIQUE (Pname),
FOREIGN KEY (Dnum) **REFERENCES** DEPARTMENT(Dnumber));

5.

CREATE TABLE WORKS_ON

(Essn	CHAR(9)	NOT NULL,
Pno	INT	NOT NULL,
Hours	DECIMAL(3,1)	NOT NULL,

PRIMARY KEY (Essn, Pno),
FOREIGN KEY (Essn) **REFERENCES** EMPLOYEE(Ssn),
FOREIGN KEY (Pno) **REFERENCES** PROJECT(Pnumber));

6.

CREATE TABLE DEPENDENT

(Essn	CHAR(9)	NOT NULL,
Dependent_name	VARCHAR(15)	NOT NULL,
Sex	CHAR,	
Bdate	DATE,	
Relationship	VARCHAR(8),	

PRIMARY KEY (Essn, Dependent_name),
FOREIGN KEY (Essn) **REFERENCES** EMPLOYEE(Ssn));

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Using the Company Database schema you created (which includes the following tables: EMPLOYEE, DEPARTMENT, DEPT_LOCATIONS, PROJECT, WORKS_ON, and DEPENDENT), write the appropriate INSERT statements to insert the data shown in the figures below into the corresponding tables.

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

DEPARTMENT

Dname	Dnumber	Mgr_ssn	Mgr_start_date
Research	5	333445555	1988-05-22
Administration	4	987654321	1995-01-01
Headquarters	1	888665555	1981-06-19

DEPT_LOCATIONS

Dnumber	Dlocation
1	Houston
4	Stafford
5	Bellaire
5	Sugarland
5	Houston

DEPENDENT

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EMPLOYEE table

```

INSERT INTO EMPLOYEE (Fname, Minit, Lname, Ssn, Bdate, Address, Sex, Salary,
Super_ssn, Dno) VALUES
('John', 'B', 'Smith', '123456789', '1965-01-09', '731 Fondren, Houston, TX', 'M', 30000,
'333445555', 5),
('Franklin', 'T', 'Wong', '333445555', '1955-12-08', '638 Voss, Houston, TX', 'M', 40000,
'888665555', 5),
('Alicia', 'J', 'Zelaya', '999887777', '1968-01-19', '3321 Castle, Spring, TX', 'F', 25000,
'987654321', 4),
('Jennifer', 'S', 'Wallace', '987654321', '1941-06-20', '291 Berry, Bellaire, TX', 'F', 43000,
'888665555', 4),
('Ramesh', 'K', 'Narayan', '666884444', '1962-09-15', '975 Fire Oak, Humble, TX', 'M', 38000,
'333445555', 5),
('Joyce', 'A', 'English', '453454353', '1972-07-31', '5631 Rice, Houston, TX', 'F', 25000,
'333445555', 5),
('Ahmad', 'V', 'Jabbar', '987987987', '1969-03-29', '980 Dallas, Houston, TX', 'M', 25000,
'987654321', 4),
('James', 'E', 'Borg', '888665555', '1937-11-10', '450 Stone, Houston, TX', 'M', 55000, NULL,
1);
  
```

DEPARTMENT table

```

INSERT INTO DEPARTMENT (Dname, Dnumber, Mgr_ssn, Mgr_start_date) VALUES
('Research', 5, '333445555', '1988-05-22'),
('Administration', 4, '987654321', '1995-01-01'),
('Headquarters', 1, '888665555', '1981-06-19');
  
```

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DEPT_LOCATIONS table**INSERT INTO DEPT_LOCATIONS (Dnumber, Dlocation) VALUES**

(1, 'Houston'),
(4, 'Stafford'),
(5, 'Bellaire'),
(5, 'Sugarland'),
(5, 'Houston');

PROJECT table**INSERT INTO PROJECT (Pname, Pnumber, Plocation, Dnum) VALUES**

('ProductX', 1, 'Bellaire', 5),
('ProductY', 2, 'Sugarland', 5),
('ProductZ', 3, 'Houston', 5),
('Computerization', 10, 'Stafford', 4),
('Reorganization', 20, 'Houston', 1),
('Newbenefits', 30, 'Stafford', 4);

WORKS_ON table**INSERT INTO WORKS_ON (Essn, Pno, Hours) VALUES**

('123456789', 1, 32.5),
('123456789', 2, 7.5),
('666884444', 3, 40.0),
('453454353', 1, 20.0),
('453454353', 2, 20.0),
('333445555', 2, 10.0),
('333445555', 3, 10.0),
('333445555', 10, 10.0),
('333445555', 20, 10.0),
('999887777', 30, 30.0),
('999887777', 10, 10.0),
('987987987', 10, 35.0),
('987987987', 30, 5.0),
('987654321', 30, 20.0),
('987654321', 20, 15.0),
('888665555', 20, NULL);

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DEPENDENT table

```

INSERT INTO DEPENDENT (Essn, Dependent_name, Sex, Bdate, Relationship) VALUES
('333445555', 'Alice', 'F', '1986-04-05', 'Daughter'),
('333445555', 'Theodore', 'M', '1983-10-25', 'Son'),
('333445555', 'Joy', 'F', '1958-05-03', 'Spouse'),
('987654321', 'Abner', 'M', '1942-02-28', 'Spouse'),
('123456789', 'Michael', 'M', '1988-01-04', 'Son'),
('123456789', 'Alice', 'F', '1988-12-30', 'Daughter'),
('123456789', 'Elizabeth', 'F', '1967-05-05', 'Spouse');
  
```

Queries:

1. Display all information about employees who work in department number 5.

```

SELECT *
FROM EMPLOYEE
WHERE Dno = 5;
  
```

	Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
1	John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000.00	333445555	5
2	Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000.00	888665555	5
3	Joyce	A	English	453454353	1972-07-31	5631 Rice, Houston, TX	F	25000.00	333445555	5
4	Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000.00	333445555	5

2. List the names of all projects located in 'Houston'.

```

SELECT Pname
FROM PROJECT
WHERE Plocation = 'Houston';
  
```

	Pname
1	ProductZ
2	Reorganization

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3. Find all employees whose salary is greater than 40000.

```

SELECT Fname, Lname, Salary
FROM EMPLOYEE
WHERE Salary > 40000;
  
```

	Fname	Lname	Salary
1	James	Borg	55000.00
2	Jennifer	Wallace	43000.00

4. Show all dependents who are male.

```

SELECT *
FROM DEPENDENT
WHERE Sex = 'M';
  
```

	Essn	Dependent_name	Sex	Bdate	Relationship
1	123456789	Michael	M	1988-01-04	Son
2	333445555	Theodore	M	1983-10-25	Son
3	987654321	Abner	M	1942-02-28	Spouse

5. Display all projects that have a project number greater than 10.

```

SELECT *
FROM PROJECT
WHERE Pnumber > 10;
  
```

	Pname	Pnumber	Plocation	Dnum
1	Reorganization	20	Houston	1
2	Newbenefits	30	Stafford	4

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6. List all dependents born before the year 1990.

```

SELECT *
FROM DEPENDENT
WHERE Bdate < '1990-01-01';
  
```

	Essn	Dependent_name	Sex	Bdate	Relationship
1	123456789	Alice	F	1988-12-30	Daughter
2	123456789	Elizabeth	F	1967-05-05	Spouse
3	123456789	Michael	M	1988-01-04	Son
4	333445555	Alice	F	1986-04-05	Daughter
5	333445555	Joy	F	1958-05-03	Spouse
6	333445555	Theodore	M	1983-10-25	Son
7	987654321	Abner	M	1942-02-28	Spouse

7. Find all records in WORKS_ON table where hours is greater than or equal to 30.

```

SELECT *
FROM WORKS_ON
WHERE Hours >= 30;
  
```

	Essn	Pno	Hours
1	123456789	1	32.5
2	666884444	3	40.0
3	987987987	10	35.0
4	999887777	30	30.0

8. Select employees who have a middle initial of 'B'.

```

SELECT *
FROM EMPLOYEE
WHERE Minit = 'B';
  
```

	Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
1	John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000.00	333445555	5

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9. List the names of all employees who are female.

```
SELECT Fname, Lname  
FROM EMPLOYEE  
WHERE Sex = 'F';
```

	Fname	Lname
1	Joyce	English
2	Jennifer	Wallace
3	Alicia	Zelaya

10. Find all dependents with the relationship 'Spouse'.

```
SELECT *  
FROM DEPENDENT  
WHERE Relationship = 'Spouse';
```

	Essn	Dependent_name	Sex	Bdate	Relationship
1	123456789	Elizabeth	F	1967-05-05	Spouse
2	333445555	Joy	F	1958-05-03	Spouse
3	987654321	Abner	M	1942-02-28	Spouse